Port power: VBAT -> MT3608 boost 5.1 V -> one high-side switch per port -> B0505S (channel sheet). EN high = port on.

Waveshare ESP32-S3-Touch-LCD-2 on two 1x14 sockets

J1
WS header 1-14

J2
WS header 15-28

J3
EXT
Spare GPIO 2/4/6 + I2C (GPIO47/48) on J3 for future add-ons (buzzer, status LEDs, expander)

C51 10u
C52 22u
C53 22u
C54 100n
D5 SS14
U5 MT3608
L1 22u

R51 75k
R52 10k
R53 100k

D51 1N4148W
D52 1N4148W
D53 1N4148W
D54 1N4148W

JP11 EN1 dflt
JP12 EN2 dflt
JP13 EN3 dflt
JP14 EN4 dflt

R61 10k
R62 10k
R63 10k
R64 10k

Q11 2N7002
Q12 AO3401A
Q21 2N7002
Q22 AO3401A
Q31 2N7002
Q32 AO3401A
Q41 2N7002
Q42 AO3401A

R71 100k
R72 100k
R73 100k
R74 100k

EN1..EN4 GPIO16/17/18/10 (10k pull-down via JP11..JP14; bridge 2-3 for always-on without firmware support)

CH1 — File: channel.kicad_sch
CH2 — File: channel.kicad_sch
CH3 — File: channel.kicad_sch
CH4 — File: channel.kicad_sch

H1 M2 Waveshare
H2 M2 Waveshare
H3 M2 Waveshare
H4 M2 Waveshare
H5 M2.5 enclosure
H6 M2.5 enclosure
H7 M2.5 enclosure
H8 M2.5 enclosure

power flags: GND / 3V3 / VBAT come from the Waveshare module, +5V from the boost converter

generated by hardware/carrier/generate.py

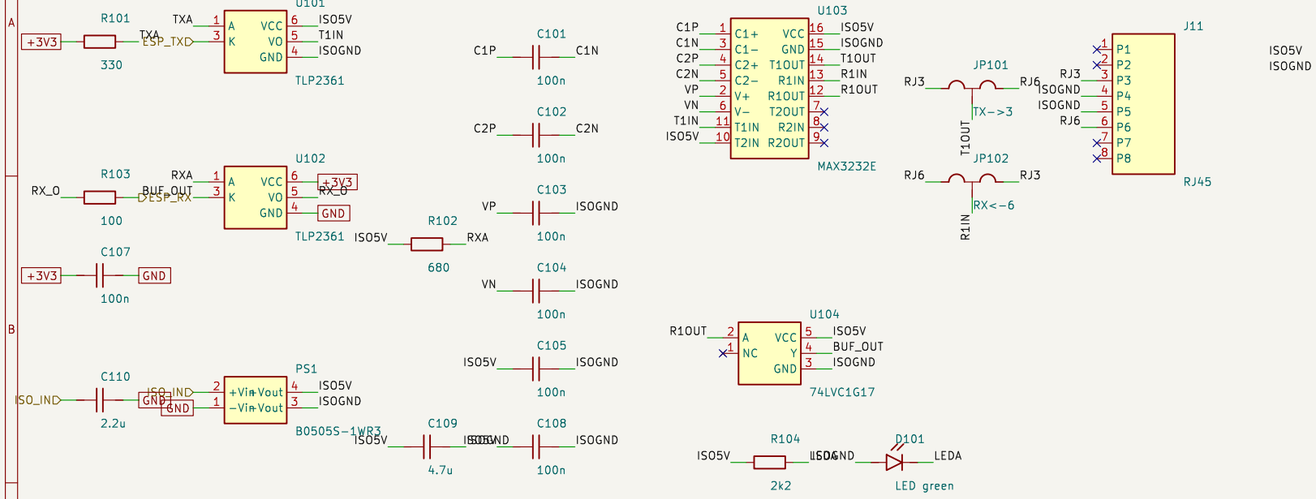
Sheet: /
File: rs232_carrier.kicad_sch

Title: RS232 isolation carrier — Waveshare ESP32-S3-Touch-LCD-2

Size: A3	Date: 2026-10-27	**Rev: 0.1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/5

ESP side (3V3 / GND) ---- isolation barrier ---- optocouplers + DC-DC ----

isolated side (ISO5V / ISOGND), RJ45 Cisco pinout: 3 TX out, 6 RX in, 4+5 GND



LED off while the line idles (UART mark = LED off) -> no quiescent LED current.

JP01/JP02 bridged 1-2 = Cisco console pinout (rollover cable to a switch). Bridge 2-3 on both to swap pins 3 and 6.

generated by hardware/carrier/generate.py

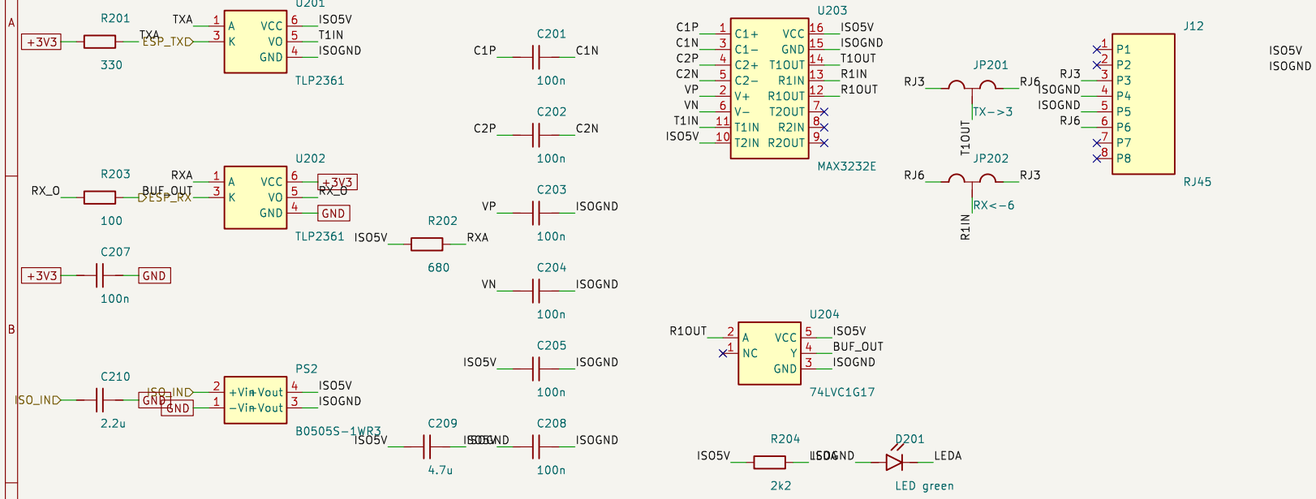
Sheet: /CH1/
File: channel.kicad_sch

Title: Isolated RS232 port (one of four)

Size: A3	Date: 2026-10-27	**Rev: 0.1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 2/5

ESP side (3V3 / GND) ---- isolation barrier ---- optocouplers + DC-DC ----

isolated side (ISO5V / ISOGND), RJ45 Cisco pinout: 3 TX out, 6 RX in, 4+5 GND



LED off while the line idles (UART mark = LED off) -> no quiescent LED current.

JP01/JP02 bridged 1-2 = Cisco console pinout (rollover cable to a switch). Bridge 2-3 on both to swap pins 3 and 6.

generated by hardware/carrier/generate.py

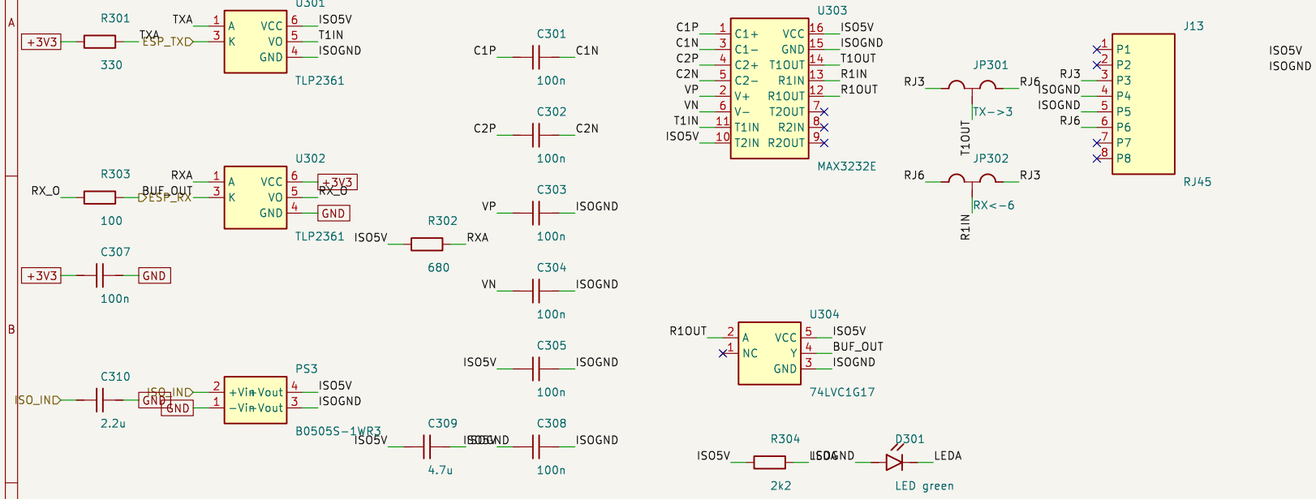
Sheet: /CH2/
File: channel.kicad_sch

Title: Isolated RS232 port (one of four)

Size: A3	Date: 2026-10-27	**Rev: 0.1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 3/5

ESP side (3V3 / GND) ---- isolation barrier ---- optocouplers + DC-DC ----

isolated side (ISO5V / ISOGND), RJ45 Cisco pinout: 3 TX out, 6 RX in, 4+5 GND



LED off while the line idles (UART mark = LED off) -> no quiescent LED current.

JP01/JP02 bridged 1-2 = Cisco console pinout (rollover cable to a switch). Bridge 2-3 on both to swap pins 3 and 6.

generated by hardware/carrier/generate.py

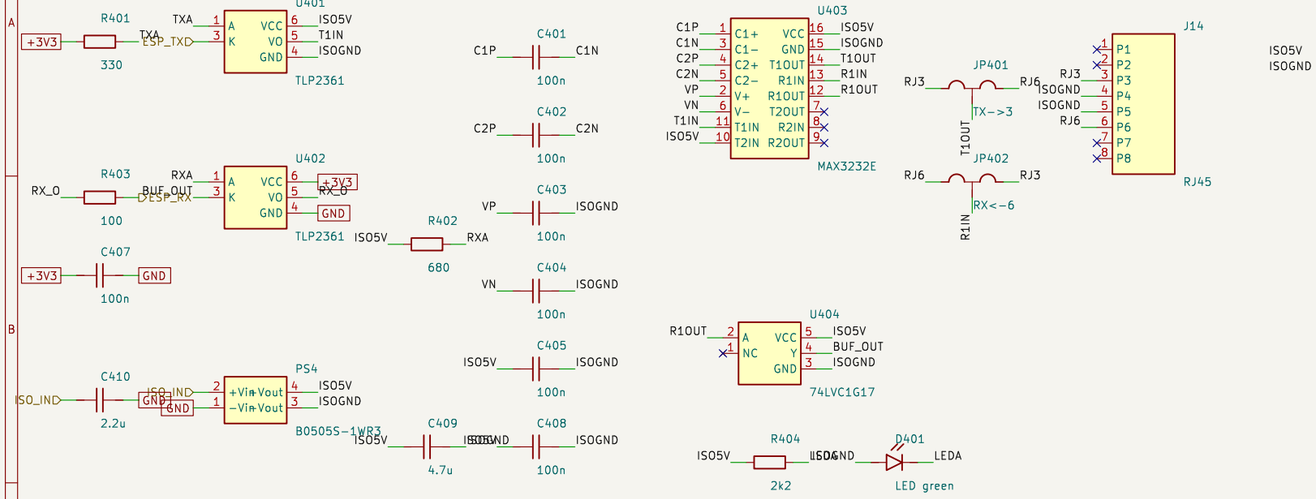
Sheet: /CH3/
File: channel.kicad_sch

Title: Isolated RS232 port (one of four)

Size: A3	Date: 2026-10-27	**Rev: 0.1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 4/5

ESP side (3V3 / GND) ---- isolation barrier ---- optocouplers + DC-DC ----

isolated side (ISO5V / ISOGND), RJ45 Cisco pinout: 3 TX out, 6 RX in, 4+5 GND



LED off while the line idles (UART mark = LED off) -> no quiescent LED current.

JP01/JP02 bridged 1-2 = Cisco console pinout (rollover cable to a switch). Bridge 2-3 on both to swap pins 3 and 6.

generated by hardware/carrier/generate.py

Sheet: /CH4/
File: channel.kicad_sch

Title: Isolated RS232 port (one of four)

Size: A3	Date: 2026-10-27	**Rev: 0.1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 5/5